

Epidemiology of the metabolic syndrome in the USA.

The metabolic syndrome is a common complex entity that has emerged as a worldwide epidemic and major public health care concern with a prevalence of approximately 25% in the United States. There have been a number of different definitions of the metabolic syndrome but all center around the metabolic abnormalities of central obesity, hypertension, decreased high-density lipoproteins and elevated triglycerides with insulin resistance as the uniting physiologic factor. The importance of the metabolic syndrome is not just related to its high prevalence rate but also because it predicts the development of diabetes and cardiovascular disease. Nonalcoholic fatty liver disease is now recognized to be the hepatic component of the metabolic syndrome, which along with its individual components - particularly diabetes and elevated triglycerides, are the major risk factors for the development of nonalcoholic steatohepatitis (NASH); the most severe form of nonalcoholic fatty liver disease. NASH may progress to cirrhosis, hepatocellular carcinoma, and liver failure. It is currently the third most common cause for liver transplantation and is projected to be the leading cause for liver transplantation in 2020. Weight loss (via diet or bariatric surgery) and vitamin E have recently been demonstrated to be effective treatments of NASH. Although these and other agents may prove to be effective treatments for NASH, the most effective therapeutic strategy would be early screening and intervention to prevent the development of insulin resistance and oxidative stress at a societal level. © 2011 The Author. Journal of Digestive Diseases © 2011 Chinese Medical Association Shanghai Branch, Chinese Society of Gastroenterology, Renji Hospital Affiliated to Shanghai Jiaotong University School of Medicine and Blackwell Publishing Asia Pty Ltd.